

Sec. VI-E, (98)–(105): the onset under a perturbation

drop-in main text; the measured sizes live in Sec. VIII-E

Because the measured angular rate $y(t)$ can be regarded as the nominal response perturbed by the unmodelled forcing ρ , the minimiser selects $\hat{t}_c$ by minimising the two-segment cost over the record. Both segments enter, and both share the pre-onset level C :

$$J(t_c) = \int_{t_0}^{t_c} (y(t) - C)^2 dt + \int_{t_c}^{t_{\text{end}}} \left(y(t) - (\hat{\omega}(t; t_c) + C) \right)^2 dt. \quad (98)$$

Write the recorded rate as the true pre-onset level, the nominal response from the true onset, and a deviation:

$$y(t) = b + \hat{\omega}(t; t_c^*) + e_\omega(t), \quad \hat{t}_c = t_c^* + \delta t_c. \quad (99)$$

Here b is the rate the vehicle actually holds before the excitation, zero under the onset conditions. The rate is taken from the navigation filter, whose delta-angle bias states remove the gyro bias, so b represents residual pre-excitation motion rather than sensor bias; C is its estimate, formed over the pre-onset segment. Note also that e_ω is anchored at the *true* onset t_c^* : once a run is recorded it is a fixed function of t , and it carries no dependence on the optimisation variable.

A first-order expansion of the estimated rate gives

$$\hat{\omega}(t; \hat{t}_c) \simeq \hat{\omega}(t; t_c^*) + \delta t_c \chi(t) \quad (100)$$

with the sensitivity

$$\chi(t) \triangleq \left. \frac{\partial \hat{\omega}(t; t_c)}{\partial t_c} \right|_{t_c=t_c^*} = -C_1 C_2 \sinh C_2 (t - t_c^*). \quad (101)$$

Two features of (98) make the differentiation clean, and both are worth stating because each looks like an omission otherwise.

First, t_c is the upper limit of one integral and the lower limit of the other, so the Leibniz rule produces two boundary terms. They cancel identically. The rise contributes $C_1(\cosh 0 - 1) = 0$ at $t = t_c$, so the two segments agree there and both terms equal $(y(t_c) - C)^2$ with opposite signs. Nothing is discarded, and the cancellation is exact rather than asymptotic.

Second, substituting (99) and (100), the post-onset residual is

$$y(t) - \hat{\omega}(t; \hat{t}_c) - C = e_\omega(t) - \delta t_c \chi(t) - (C - b), \quad (102)$$

so C enters only through the baseline *error* $C - b$, not through its own size. A large but correctly estimated pre-onset level is harmless.

Differentiating (98) with respect to δt_c — the same operator as $\partial/\partial t_c$ by (99), but the variable in which the linearised cost is an exact quadratic — and setting the result to zero gives, when the pre-onset fit is unbiased,

$$\frac{\partial J}{\partial \delta t_c} = -2 \int_{t_c}^{t_{\text{end}}} (e_\omega(t) - \delta t_c \chi(t)) \chi(t) dt = 0, \quad (103)$$

which implies

$$\delta t_c = \frac{\langle e_\omega, \chi \rangle}{\|\chi\|^2}. \quad (104)$$

Here $\langle a, b \rangle$ denotes the L^2 inner product over the fitting window $[t_c^*, t_{\text{end}}]$, i.e. $\langle a, b \rangle \triangleq \int_{t_c^*}^{t_{\text{end}}} a(t)b(t) dt$, and $\|b\|^2 \triangleq \langle b, b \rangle$. The shift δt_c is well defined because the maximum-angle event occurs after the onset, $\tau_{\text{end}} > 0$, so $\sinh C_2 \tau$ does not vanish identically and $\|\chi\|^2 > 0$.

The baseline error, retained. Keeping the last term of (102) rather than assuming it away replaces (104) by

$$\delta t_c = \frac{\langle e_\omega, \chi \rangle}{\|\chi\|^2} - (C - b) \frac{\langle 1, \chi \rangle}{\|\chi\|^2}, \quad (105)$$

and the second term does not vanish: χ is of one sign, so $\langle 1, \chi \rangle = -C_1(\cosh x - 1)$. Evaluating it with $\|\chi\|^2 = C_1^2 C_2 [\frac{1}{4} \sinh 2x - \frac{1}{2}x]$ and carrying it through $\Delta M_{\text{crit}} = M \delta t_c$, the amplitude C_1 cancels as it does elsewhere and

$$\Delta M_{\text{crit}, C} = W z_{CoM} (C - b) \frac{\cosh x - 1}{C_2 [\frac{1}{4} \sinh 2x - \frac{1}{2}x]}. \quad (106)$$

Over the operating box of Sec. VI-D the gain of (106) runs from 0.042 N m per rad/s at the slowest ramp to 0.351 at the fastest, the fast end being worse because the window is shorter. What the fitted baselines actually are, and how far the identified offset moves when C is pinned to zero instead of fitted, are measurements and are reported with the realised budget in Sec. VIII-E.